KERALA POLYTECHNIC | POLYMER TECHNOLOGY

CAD Lab I

Course Code: 2099 Revision: 2021 Semester: 2

Download Official Syllabus PDF

Course Overview

Teaching Scheme

- **L:T:P :** 0:0:3
- **Periods/Week :** 3
- **Total Periods :** 45
- **Credits :** No Credit

Course Objectives

- Hands-on training on 2D drawing using AutoCAD.
- Competency in industrial 2D drawing modification.
- Platform for mastering AutoCAD 3D.

Prerequisites

Basic Mathematics (Mathematics I) and Engineering Graphics (Orthographic Projections).

Course Outcomes (CO)

CO No.	Description	Duration	Cognitive Level
CO1	Explain computer aided drafting and various commands used in AutoCAD	9 Hours	Applying
CO2	Draw 2D drawings and sectional views in AutoCAD	17 Hours	Applying
CO3	Perform dimensioning and plotting in AutoCAD	12 Hours	Applying
CO4	Develop Isometric drawings of simple objects in AutoCAD	5 Hours	Applying

Module 1: AutoCAD Basics & Commands

9 Hours CO1

1.1 Introduction to CAD

Computer-Aided Drafting (CAD) is the use of computer systems to assist in the creation, modification, analysis, or optimization of a design. It replaces manual drafting with an automated process.

Advantages over Manual Drafting:

- **Speed and Productivity:** Drawings can be created much faster.
- **Accuracy:** Extremely high precision (up to 14 decimal places).
- **Ease of Editing:** Modifications like "Undo", "Erase", or "Move" are instantaneous.

- **Storage and Retrieval:** Digital files take no physical space and are easy to share.
- **Standardization:** Use of layers and blocks ensures uniformity.

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1.2 Coordinate Systems

AutoCAD uses the Cartesian Coordinate System. There are four main ways to enter points:

1. **Absolute Coordinate System:** Points are measured from the origin (0,0). Format: `X,Y`
2. **Relative Rectangular Coordinate System:** Points are measured from the last point. Format: `@X,Y`
3. **Relative Polar Coordinate System:** Points are defined by distance and angle from the last point. Format: `@Distance<Angle`
4. **Direct Distance Entry:** Move the cursor in a direction and type the distance.

> **Example: Drawing a square of side 50 units using Relative Coordinates**
> Start Point: 10,10
> Next Point (Right): @50,0
> Next Point (Up): @0,50
> Next Point (Left): @-50,0
> Close: @0,-50 or 'C'

1.3 Draw Commands

Essential commands for creating geometry:

- **LINE:** Creates straight line segments.

- **CIRCLE:** Options include Center-Radius, Center-Diameter, 2-Point, 3-Point.
- **ARC:** Creates curved segments.
- **RECTANGLE:** Creates a closed polyline rectangle.
- **POLYGON:** Creates equilateral closed polylines (Inscribed/Circumscribed).
- **HATCH:** Fills enclosed areas with patterns (used for sectional views).

Module 2: 2D Drawings & Sectional Views

17 Hours CO2

2.1 Modify Commands

These commands are used to edit existing objects:

Command	Function
OFFSET	Creates concentric circles or parallel lines.
TRIM	Cuts objects at a cutting edge.
FILLET	Rounds the edges of two objects.
CHAMFER	Bevels the edges of two objects.
ARRAY	Creates multiple copies in a pattern (Rectangular/Polar).
MIRROR	Creates a reflected copy of objects.

2.2 Practical Exercises

Students must perform the following drawings:

- **I-Section:** Standard structural beam cross-section.

- **Stanchion Mount:** A support base for vertical pillars.
- **Sectional Elevation of Foot Step Bearing:** Showing internal parts like the bush and shaft.
- **First & Third Angle Projection Symbols:** Essential for engineering drawing standards.

Malayalam Note: Sectional views [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] 'Hatch' [illegible] [illegible] [illegible] [illegible] [illegible].

Module 3: Dimensioning & Plotting

12 Hours CO3

3.1 Dimensioning Styles

Dimensioning is critical for manufacturing. In AutoCAD, use `DIMSTYLE` to manage:

- **Lines:** Dimension lines and extension lines.
- **Symbols and Arrows:** Arrowhead size and type.
- **Text:** Height, placement, and alignment.
- **Primary Units:** Precision and decimal separators.

3.2 Polymer Specific Drawings

As this is for Polymer Technology, specific test samples are drawn:

- **Dumbbell Tensile Sample:** Standard shape used for testing tensile strength of polymers (ASTM D638).
- **Angular Tear Sample:** Used for testing tear resistance (ASTM D624).

- **Factory Layout:** Planning the placement of injection molding machines, extruders, and raw material storage.

3.3 Plotting (Printing)

The process of converting a digital drawing to a physical print or PDF:

1. Command: `PLOT` or `CTRL+P`.
2. Select Printer/Plotter (e.g., DWG to PDF).
3. Select Paper Size (A4, A3, etc.).
4. Plot Area: Window, Extents, or Display.
5. Scale: 1:1, Fit to Paper, etc.

Module 4: Isometric Drawings

5 Hours **CO4**

4.1 Isometric Basics

Isometric drawing is a 2D representation of a 3D object where the three axes are 120° apart. In AutoCAD:

- Enable Isometric Snap: `SNAP` -> Style -> Isometric.
- Switch Planes (Top, Left, Right): Use `F5` or `ISOPLANE` command.
- **ISOCIRCLE:** Use the Ellipse command (Axis-End option) then select 'Isocircle' to draw circles in isometric view.

Isometric Checklist: 1. Grid/Snap set to Isometric.
2. Ortho Mode ON (F8).

3. F5 to toggle between planes.
4. Never use 'Circle' command; always use 'Ellipse -> Isocircle'.

Formula Bank & Calculations

```
Polar Coordinate: @Distance < Angle
```

Coordinate Conversion:
If you are at (10,10) and want to go to (50,50):
- Absolute: 50,50
- Relative Rectangular: @40,40
- Relative Polar: @56.57<45

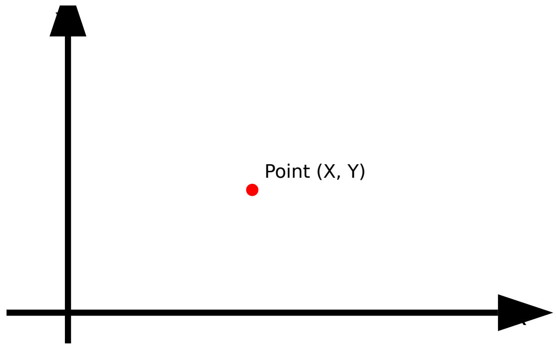
Scale Calculator

Actual Length (mm): e.g. 1000

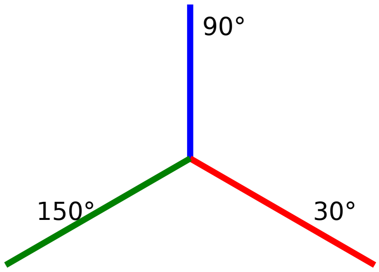
Scale (1:X): e.g. 10

Calculate Drawing Length

Visual Library



Cartesian Coordinate System in AutoCAD



Isometric Axes (30°, 90°, 150°)

Module-wise Practice Questions

Q1: What is the difference between TRIM and ERASE? (3 Marks)

ERASE: Removes the entire object from the drawing.
TRIM: Removes only a part of an object that crosses a cutting edge (boundary).

Q2: Explain the use of Layers in AutoCAD. (5 Marks)

Layers are like transparent overlays. They allow you to:
1. Organize objects by function (e.g., Hidden lines, Dimensions, Text).
2. Control visibility (Turn On/Off).
3. Lock objects to prevent accidental editing.
4. Assign colors, linetypes, and lineweights globally.

Q3: How do you draw a circle in isometric view? (3 Marks)

You cannot use the standard CIRCLE command. You must:
1. Set Snap to Isometric.
2. Enter the ELLIPSE command.

3. Choose the 'Isocircle' option.
4. Specify center and radius/diameter.

Practice Model Paper

Time: 3 Hours | Max Marks: 75 (Internal Lab Exam)

Part A: Basic Commands (20 Marks)

1. Draw a rectangle of 100x50 mm using relative coordinates. (10)

2. Create a hexagon of side 30mm using the Polygon command. (10)

Part B: 2D Drafting (30 Marks)

3. Draw the orthographic views (Front and Top) of an Index Plate as per the given dimensions. (30)

Part C: Isometric & Sectional (25 Marks)

4. Draw the Isometric view of a V-Block. (25)

Quick Revision

- **F3:** Toggle Object Snap (OSNAP).
- **F5:** Toggle Isoplanes.
- **F7:** Toggle Grid.
- **F8:** Toggle Ortho Mode.
- **Limits:** Defines the drawing area boundary.
- **Units:** Sets decimal/architectural and precision.

- **Explode:** Breaks a polyline or block into individual segments.

Glossary

GUI
: Graphical User Interface - the visual part of the software.

Object Snap
: Tool to snap to exact points like Midpoint, Endpoint, or Center.

Polyline
: A single object composed of multiple line/arc segments.

Template
: A file (.dwt) with preset layers, units, and styles.

Official Source Declaration

This content is based on the **SITTTR Kerala Revision 2021** syllabus for the course **CAD Lab I (Code: 2099)**. The syllabus is intended for the Diploma in Polymer Technology.

Official Source: www.sitttrkerala.ac.in

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- **AutoCAD Basics:** [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].
- **Coordinate Systems:** [illegible] [illegible] [illegible] Absolute (@X,Y [illegible]), Relative (@X,Y), Polar (@Dist<Angle) [illegible] [illegible] [illegible].
- **Modify Commands:** [illegible] [illegible] [illegible] [illegible] Offset ([illegible] [illegible]), Trim ([illegible] [illegible] [illegible]), Fillet ([illegible] [illegible]) [illegible] [illegible].

- **Polymer Samples:** □□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□□□□□□ □□□□□□□□□ Dumbbell tensile sample, Angular tear sample □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□.